Dr. Bailey J. Anderson

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University Training

10/2019-01/2024	**DPhil** (*PhD*) Environmental Research (NERC DTP), Department of Geography and the Environment. University of Oxford. Defence in November 2023.
09/2017-09/2019	**MPhil** in Water Science, Policy and Managemen*t*, Department of Geography and the Environment. University of Oxford
08/2013-12/2016	**Bachelor** of Arts in Geography, The University of Texas at Austin

Research Employment Since PhD

01/2024-Present	**Postdoctoral Researcher:** ETH Zurich and WSL Institute for Snow and Avalanche
10/2023-12/2023	**Research associate in computational hydrology**: University of Oxford

Selected Grants and Awards

2026-2028	Schmidt AI in Science Fellowship, awarded December 2025 (207,000 USD)
2025	ETH Studio Davos Grant (3500 USD)
2024	Swiss Society for Hydrology and Limnology Junior Promotion Fund Grant (600 USD)
2019-2023	University of Oxford Clarendon Fund Scholarship – NERC DTP (267,000 USD)
2017-2019	Marshall Scholarship (140,000 USD)
2015-2017	National Oceanic and Atmospheric Administration (NOAA) Hollings Scholar (32,000 USD)
2015	Fulbright Killam Fellowship (7,000 USD)

2022	Outstanding Reviewer for *Water Resources Research*
2019	Distinction from School of Geography and the Environment for Master's thesis

Publications

Peer Reviewed Research

1. **Anderson B.J.,** Schillinger M., Muñoz-Castro E., Tarasova L., Berghuijs W.R., Brunner M.I. (2026). How extreme are transitions in hydrology? A conditional probability approach. Environmental Research Letters.
2. Gnann S., **Anderson B.J.,** Weiler M. (*2026*). Uncertainty and non-stationarity in empirical streamflow sensitivity estimates. Hydrology and Earth System Sciences Discussions.
3. Muñoz-Castro E., **Anderson B.J.,** Astagneau P.C., Mendoza P.A., Swaim D., Brunner M.I. (*2026*). How well do hydrological models simulate streamflow extremes and drought-to-flood transitions? Hydrology and Earth System Sciences Discussions.
4. Brunner M.I., Mittermeier M., **Anderson B.J.,** Bueler D., Muñoz-Castro E. (2025). Spatially compounding drought-flood events are favored by atmospheric blocking over Europe. Water Resources Research.
5. **Anderson, B. J.,** Muñoz-Castro, E., Tallaksen, L. M., Matano, A., Götte, J., Armitage, R., Magee, E., & Brunner, M. I. (2025). What is a drought-to-flood transition? Pitfalls and recommendations for defining consecutive hydrological extreme events. Hydrology and Earth System Sciences, 29(21), 6069–6092. https://doi.org/10.5194/hess-29-6069-2025
6. **Anderson, B.J.,** Slater L.S., Rapson, J., Brunner, M.I., Dadson, S.J., Yin, J., Buechel, M. (*2025*). Interannual variability in streamflow sensitivity to precipitation is large in arid regions. Earth's Future.
7. Brunner M.I**., Anderson B.J.,** Munoz-Castro E., (*2025*). Meteorological and hydrological dry-to-wet transition events are only weakly related. Environmental Research Letters.
8. Hammond, J., **Anderson, B.,** Simeone, C., Brunner, M., Muñoz-Castro, E., Archfield, S., Magee, E., & Armitage, R. (2025). Hydrological Whiplash: Highlighting the Need for Better Understanding and Quantification of Sub-Seasonal Hydrological Extreme Transitions. Hydrological Processes, 39(3), e70113. https://doi.org/10.1002/hyp.70113
9. Berghuijs, W. R., Woods, R. A., **Anderson, B. J.,** Hemshorn de Sánchez, A. L., & Hrachowitz, M. (2025). Annual memory in the terrestrial water cycle. *Hydrology and Earth System Sciences*, *29*(5), 1319–1333. https://doi.org/10.5194/hess-29-1319-2025
10. **Anderson, B. J.,** Brunner, M. I., Slater, L. J., & Dadson, S. J. (2024). Elasticity curves describe streamflow sensitivity to precipitation across the entire flow distribution. Hydrology and Earth System Sciences Discussions, 1–27. https://doi.org/10.5194/hess-2022-407
11. **Anderson, B. J**., Slater, L. J., Dadson, S. J., Blum, A. G., & Prosdocimi, I. (2022). Statistical attribution of the influence of urban and tree cover change on streamflow: A comparison of large sample statistical approaches. Water Resources Research, 58, e2021WR030742. https://doi.org/10.1029/2021WR030742
12. Slater, L. J., **Anderson, B**., Buechel, M., Dadson, S., Han, S., Harrigan, S., Kelder, T., Kowal, K., Lees, T., Matthews, T., Murphy, C., & Wilby, R. L. (2021). Nonstationary weather and water extremes: A review of methods for their detection, attribution, and management. *Hydrology and Earth System Sciences*, *25*(7), 3897–3935. https://doi.org/10.5194/hess-25-3897-2021
13. Lees, T., Buechel, M., **Anderson, B**., Slater, L., Reece, S., Coxon, G., and Dadson, S. (2021) Benchmarking Data-Driven Rainfall-Runoff Models in Great Britain: A comparison of LSTM-based models with four lumped conceptual models. Hydrology and Earth System Sciences https://doi.org/10.5194/hess-25-5517-2021

Articles under review

1. **Anderson B.J.***, Astagneau P.C.*, Gnann S., Brunner M.I. (Under Review). Reliance on daily mean streamflow data biases inferred flood seasonality. Environmental Research Letters. *These authors contributed equally to this work.
2. Liu, Y., Wortmann, M., Hawker, L., Neal, J., Yin, J., **Anderson, B.J.,** Santos, M.S., Boothroyd, R., Nicholas, A., Smith, G.S., Ashworth, P.J., Cloke, H., Gebrechorkos, S., Parsons, D.R., Darby, S.E., Slater, L.J. (*Under Review*). Global Estimation of Bankfull River Discharge (G-QBF) Reveals Distinct Flood Recurrence Pattern across Climate Regions. Nature Water.
3. Muñoz-Castro E., **Anderson B.J.,** Brunner M.I., (Under Review). Unravelling the role of hydrometeorological conditions in streamflow drought-to-flood transitions. Water Resources Research.

Technical Reports

1. McCathran, M., J.D., **Anderson, B**., & Hermitte, S., M.P.Aff. (2015). Groundwater Production Monitoring in the State of Texas (pp. 1-92) (United States, Texas Water Development Board). Austin, Texas: Texas Water Development Board.

Teaching and Mentorship

Supervision

Ongoing	Eduardo Muñoz-Castro. **PhD Candidate** *2024-Present***.** ETH Zurich Institute for Atmospheric and Climate Science, "Consecutive drought-flood events in a warming world", Secondary Advisor.
Completed	Yanis Merzouki. **Studio Davos Intern** *2025***.** *"To what extent do drought preconditions impact flood forecast quality? "*
	Lydia Seebauer. **Bachelor's thesis** 2024-2025. ETH Zurich Bachelor's in environmental sciences, "How do specific weather patterns relate to hydrological drought severity across Europe?", First Advisor.
	Lara Schwizer*.* **Master's Thesis** ***2024-2025.*** ETH Zurich Institute for Atmospheric and Climate Science, "Which weather patterns drive hydrologic drought to flood transitions?", First Advisor.

Selected Teaching

01/2022-05/2022	**Lecturer**, Oxford University School of Geography and the Environment Designed and delivered 4-week training course on "*A practical introduction to R for environmental research*" to the Water Science, Policy and Management MSc.
01/2021-05/2022	**Teaching assistant**, Oxford University School of Geography and the Environment: Undergraduate course in Geographic Data Science
10/2020-12/2022	**Technical demonstrator and Project instructor***,* Oxford University MPLS Doctoral Training Centre and the Departments of Zoology and Plant Sciences: Statistics and Data Management

09/2020-12/2020 **Teaching assistant,** Oxford University School of Geography and the Environment: Water Science, Policy and Management MSc. Research Design and Skills

Community engagement

10/2020-07/2022	**Student Representative**: AGU Frontiers in Hydrology Conference Planning Committee
08/2020-03/2023	**Non-Executive Director**: Cultivate Oxford
10/2018-04/2019	**Master's Student Representative**: SOGE Joint Consultative Committee
04/2019-04/2020	**Women's Team Captain**: Oxford University Judo Club

Peer Reviewer: Nature, Communications Earth and Environment, Water Resources Research, Journal of Hydrology, Geophysical Research Letters, Hydrology and Earth System Science, Environmental Research Letters